

The University of the South Pacific

School of Computing, Information & Mathematical Sciences

CS412 Artificial Intelligence

Lab 3 (worth 4%)

Lab on Logistic Regression

This assessment covers the following course learning outcomes partially:

CLO 1 – Evaluate machine learning algorithms and their limitations

CLO 3 – Apply machine learning algorithms to solve real life complex computing problems.

Read the following case study to answer the given questions:

Case study is taken from: <http://nargund.com/gsu/mgs8040/lecture/LogisticRegression.Doc>

Case Study on Logistic Regression

A pharmaceutical firm that developed a particular drug for women wants to understand the characteristics that cause some of them to have an adverse reaction to a particular drug . They collect data on 15 women who had such a reaction, and 15 who did not. The variables measured are:

1. Systolic Blood Pressure
2. Cholesterol Level
3. Age of the person
4. Whether or not the woman was pregnant (1 = yes).

The dependent variable indicates if there was an adverse reaction (1 = yes).

Raw Data

BP	Cholesterol	Age	Pregnant	DrugReaction
100	150	20	0	0
120	160	16	0	0
110	150	18	0	0
100	175	25	0	0
95	250	36	0	0
110	200	56	0	0
120	180	59	0	0
150	175	45	0	0
160	185	40	0	0
125	195	20	1	0
135	190	18	1	0
165	200	25	1	0
145	175	30	1	0
120	180	28	1	0

100	180	21	1	0
100	160	19	1	1
95	250	18	1	1
120	200	30	1	1
125	240	29	1	1
130	172	30	1	1
120	130	35	1	1
120	140	38	1	1
125	160	32	1	1
115	185	40	1	1
150	195	65	0	1
130	175	72	0	1
170	200	56	0	1
145	210	58	0	1
180	200	81	0	1
140	190	73	0	1

Do the following:

1. Create logistic regression algorithm given in the lecture preferably in Matlab.
2. The class labels can be +1 or -1.
3. Make training (60%) and testing data (40%) sets randomly from the raw data. Validation for overfitting is not required in this lab. Train your algorithm on the given data set. Determine the appropriate parameters and find out the final weight vector?
4. Test your algorithm and do the prediction.
5. What is your classification accuracy?

BOK	Unsatisfactory (0%-49%)	Satisfactory (50% - 75%)	Good (76% - 100%)	Marks Allocated	% Marks Attained
Abstraction & Design	Unsatisfactory II. Irrelevant prototyping VIII. Poor architecture and design makes future updates hard to incorporate.	Satisfactory I. Analysis of requirements such as risk analysis & mitigation, prototyping, identification of true/false requirements etc. III. all relevant constraints and requirements are mentioned VI. Appropriate choice of the architecture and design	Good I. All satisfactory and ... II. Choice of architecture and design is based on robustness, scalability, evolution and long term management. III. Innovative or creative design of algorithms (efficiency, structure and data structures)	1	
Programming	Unsatisfactory I. Plagiarism II. Poor indentation, hard to read and follow the code III. Lots of bugs and/or errors IV. Program produces unexpected output VII. Hard coding of data in the program. VIII. Poor structure	Satisfactory I. Able to write a complex code for an open ended problem II. Use of highest standard of programming practices such as modularity with appropriate architecture, low coupling, high	Good I. All satisfactory and... demonstrate very good programming skills.	1	

		cohesion, standard docs (eg. Javadoc) etc. III. Appropriate testing is done.			
Data & Information Management	Unsatisfactory VII. Very limited or undesirable data has been collected IX. Poor or incorrect inference is derived from the collected/given data	Satisfactory VII. Appropriate data has been collected IX. logical inference is derived from the collected/given data	Good I. All satisfactory and III. Extensive analysis is done on the collected/given data. IV. Exceptional Inference is derived from the data.	2	
Sub Total & comments				4	